

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & 0 \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$

$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$

$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$

$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$

$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{zzz} = \mathbb{M}_{yz}^{(2)} + \mathbb{T}_x^{(3)}$$

$$T_{xyy} = \mathbb{M}_{yz}^{(2)} - \mathbb{T}_x^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzy} & 0 & 0 \\ T_{xyz} & T_{yyy} & T_{zzy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxy} \\ 0 & 0 & 0 & 0 & T_{zxy} & 0 \end{bmatrix}$$

$$T_{xyz} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)}$$

$$T_{yyy} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zzy} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzy} & 0 & 0 \\ -T_{xyz} & -T_{yyy} & -T_{zzy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxy} \\ 0 & 0 & 0 & 0 & -T_{zxy} & 0 \end{bmatrix}$$

$$T_{xyz} = -4\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{yyy} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{zzy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{zxy} = -\mathbb{M}_x^{(1)} - 4\mathbb{M}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxy} \\ 0 & T_{zxy} & 0 \end{bmatrix}$$

$$T_{zxy} = 2\mathbb{T}_{yz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxy} \\ 0 & -T_{zxy} & 0 \end{bmatrix}$$

$$T_{zxy} = -\mathbb{M}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xyz} & 0 & 0 \\ T_{yyy} & 0 & 0 \\ T_{zzy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zxy} \\ 0 & T_{xyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} \\
T_{yyyy} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzyz} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zxxxy} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{yz}^{(5)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxxy} \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{zyyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{zxxxy} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & T_{yyyz} & 0 & 0 & T_{yyzy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzyz} & 0 & 0 & T_{zzzy} & 0 & 0 \\ T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxxy} & 0 & 0 & T_{zxyx} \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{xxzy} &= -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yyyy} &= 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzyz} &= -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yzxx} &= 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)} \\
T_{yzyy} &= -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{yzzz} &= 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxxy} &= -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{zxyx} &= -\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)} \\
T_{xyxz} &= \mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxxy} & 0 & 0 & T_{zxyx} \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\
T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\
T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\
T_{zxxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\
T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\
T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & 0 \\
0 & 0 & T_{yyyyz} \\
0 & T_{zzzy} & 0 \\
0 & T_{yyzy} & 0 \\
0 & 0 & 0 \\
0 & 0 & T_{xxyz} \\
0 & 0 & T_{yzzz} \\
0 & T_{zxxxy} & 0 \\
0 & 0 & 0 \\
-\frac{T_{yyyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxxy} & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
T_{yyyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{xxyz} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\
T_{zxxxy} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}
\end{aligned}$$